

AICE2005 - Systematic Design

Use Cases and Personas

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Week Overview

Week 2 | Analysing systems: requirements

Lecture 1: Functional and Non-Functional Requirements

What must the system do, and how well must it do it?

Lecture 2: Gathering and Validating Requirements

How do we find out what stakeholders need, and check we got it right?

Lecture 3: Use Cases and Personas

How do we capture users and their goals as scenarios?

Record your attendance



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Recap: Lectures 1 and 2

- **Lecture 1:** FRs, NFRs and how to make them measurable and unambiguous
- **Lecture 2:** eliciting requirements from stakeholders and validating them
- **Open problem:** lists of "shall" statements can still miss how real people work

Lecture 3 Overview

- Why requirements still fail
- User-centred design and use cases
- Use case anatomy, levels and a worked example
- Personas and how to build them
- Integrating and critically evaluating both

Learning Objectives:

- Define and differentiate use cases and personas as complementary techniques
- Construct well-formed use case specifications and persona profiles from research data
- Integrate use cases and personas into systems analysis workflows
- Critically evaluate the strengths, limitations and contexts of these methods

Why Requirements Still Fail

Persistent risk: industry surveys continue to cite incomplete requirements and limited user involvement as leading causes of project failure.

1. Experienced practitioners warn that requirements specification is often the hardest, and most dangerous, stage of systems development.
2. Traditional elicitation (interviews, surveys, document reviews) regularly produces abstract statements that ignore lived user context.

Use cases and personas inject narrative, empathy and testable structure into socio-technical requirements.

User-Centred Design Foundations

Core principles

- Shift design priorities from technology capabilities to human needs and contexts.
- Base design decisions on an explicit understanding of users, tasks and environments.
- Iterate solutions with continuous user participation and evaluation.

Systems lens

- Systems thinking highlights multiple stakeholder perspectives in socio-technical change.
- Personas humanise those viewpoints within complex ecosystems.
- Use cases map the interactions stakeholders have at the system boundary, helping engineers reason about feedback loops and constraints.

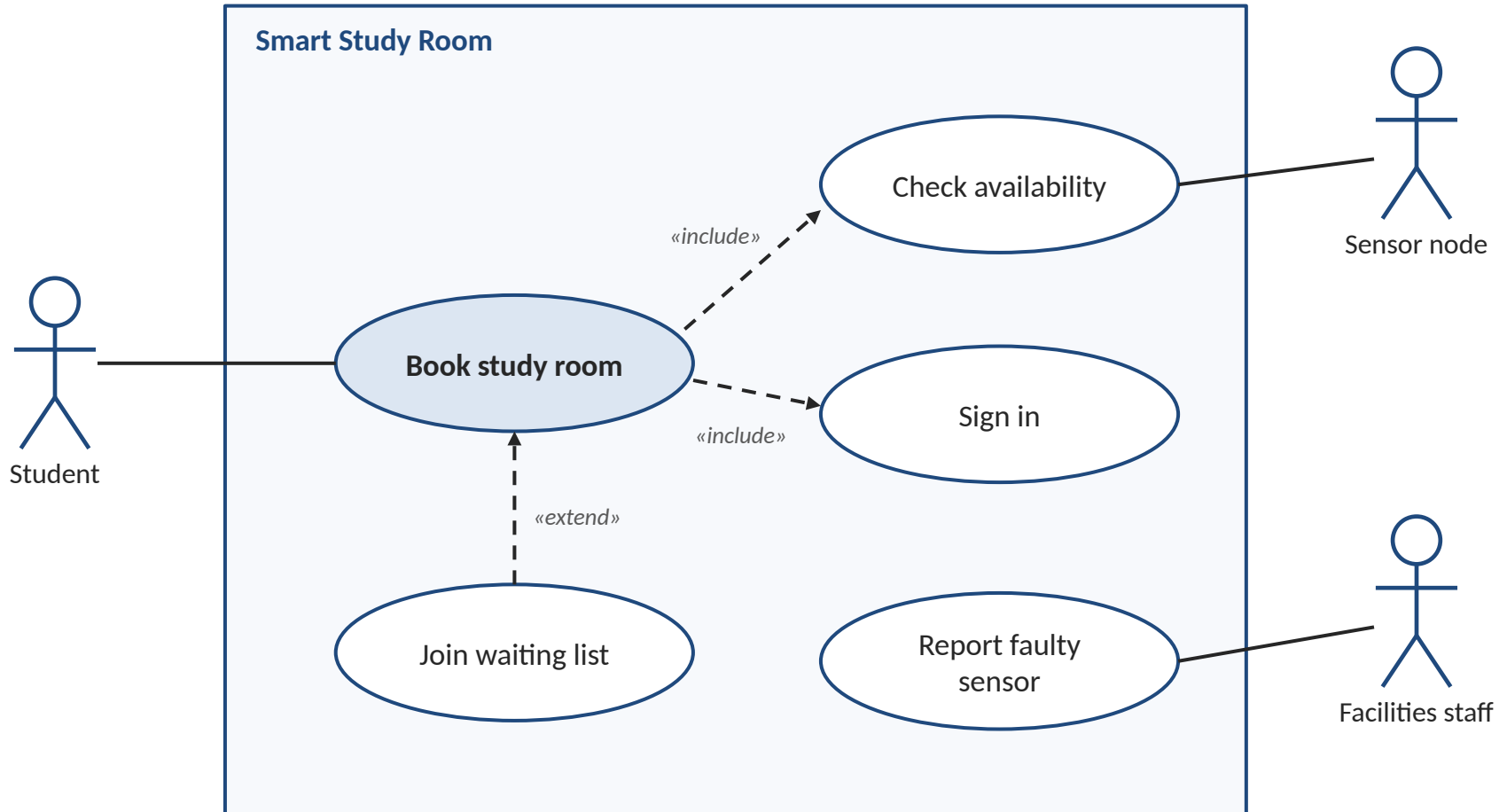
Human-centred design: ISO 9241-210:2019

Use Cases: What and Why

- 1 **Describe interactions** between an actor and the system that accomplish a goal.
- 2 **A contract for behaviour** stating what the system will do, without prescribing how.
- 3 **A bridge** between business and technical audiences, because they sit at the system boundary.

Key terms: **actor** (who or what acts), **goal** (what they want), **system boundary** (what is inside the system we are building).

A Use Case Diagram: Smart Study Room



«**include**»: always part of the goal. «**extend**»: optional or exceptional behaviour. Keep diagrams high-level; put detailed logic in text.

Use Case Anatomy

1 Name and identifier

verb + noun, e.g. UC-03 Book study room

2 Actors and stakeholders

who acts, and who else cares

3 Preconditions and triggers

what must be true first; what starts it

4 Main success scenario

numbered steps: actor intention, system response

5 Alternative flows and exceptions

what if: errors and other choices

6 Postconditions and business rules

what is true afterwards; rules that always apply

Describe intentions ("student selects a time"), not clicks ("student clicks the blue button").

Use Case Levels

Summary

cloud, kite

Strategic, multi-session goals. Give context but need decomposing.

e.g. Manage campus study space

User goal

sea level

One actor, one sitting, clear value. The primary focus for specifications.

e.g. Book a study room

Subfunction

fish

Supporting steps, reused by several use cases.

e.g. Check availability, Sign in

Too low (clam): "Type the student ID" is a step inside another use case, not a use case of its own.

Worked Example: Book a Study Room

Context

- **Primary actor:** student
- **Stakeholders:** facilities, timetabling
- **Precondition:** student is signed in
- **Trigger:** student selects "Book a room"

Main success scenario

1. System lists rooms and current availability.
2. Student selects a room and time slot.
3. System checks the booking rules.
4. System presents a summary.
5. Student confirms.
6. System records the booking and marks the slot taken.
7. System sends a confirmation.

Alternative flows

- No room free: offer the waiting list.
- Outside opening hours: show error, return to step 2.
- Sensor offline: show "status unknown", allow booking.

Business rules: maximum 2 hours per booking; 3 bookings per week; cancel at least 15 minutes before the start.

Let's think and reflect

Exercise: Which level?



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Summary, user goal, subfunction, or too low to be a use case?

1. Manage campus study space.
2. Book a study room.
3. Validate a booking against the rules.
4. Cancel a booking.
5. Reduce energy use across all campus buildings.
6. Type the student ID into the form.

Which level? Discussion

#	Goal	Level	Why
1	Manage campus study space	Summary	Many user goals over weeks
2	Book a study room	User goal	One sitting, clear value
3	Validate a booking	Subfunction	Supports booking and changing
4	Cancel a booking	User goal	One sitting; frees the room
5	Reduce campus energy use	Summary	Strategic, many systems
6	Type the student ID	Too low	A step in another use case

User-goal test: can the actor walk away satisfied after one sitting? If it is only a piece of that, it is a subfunction; if it spans days, it is a summary.

Personas: What and Why

A persona is a fictional but evidence-based profile of a group of typical users, used to make design decisions about real people.

Use cases: behaviour

What the system does for an actor to reach a goal. Text, testable.

Personas: context

Who the users are and why they act: goals, frustrations, situation.

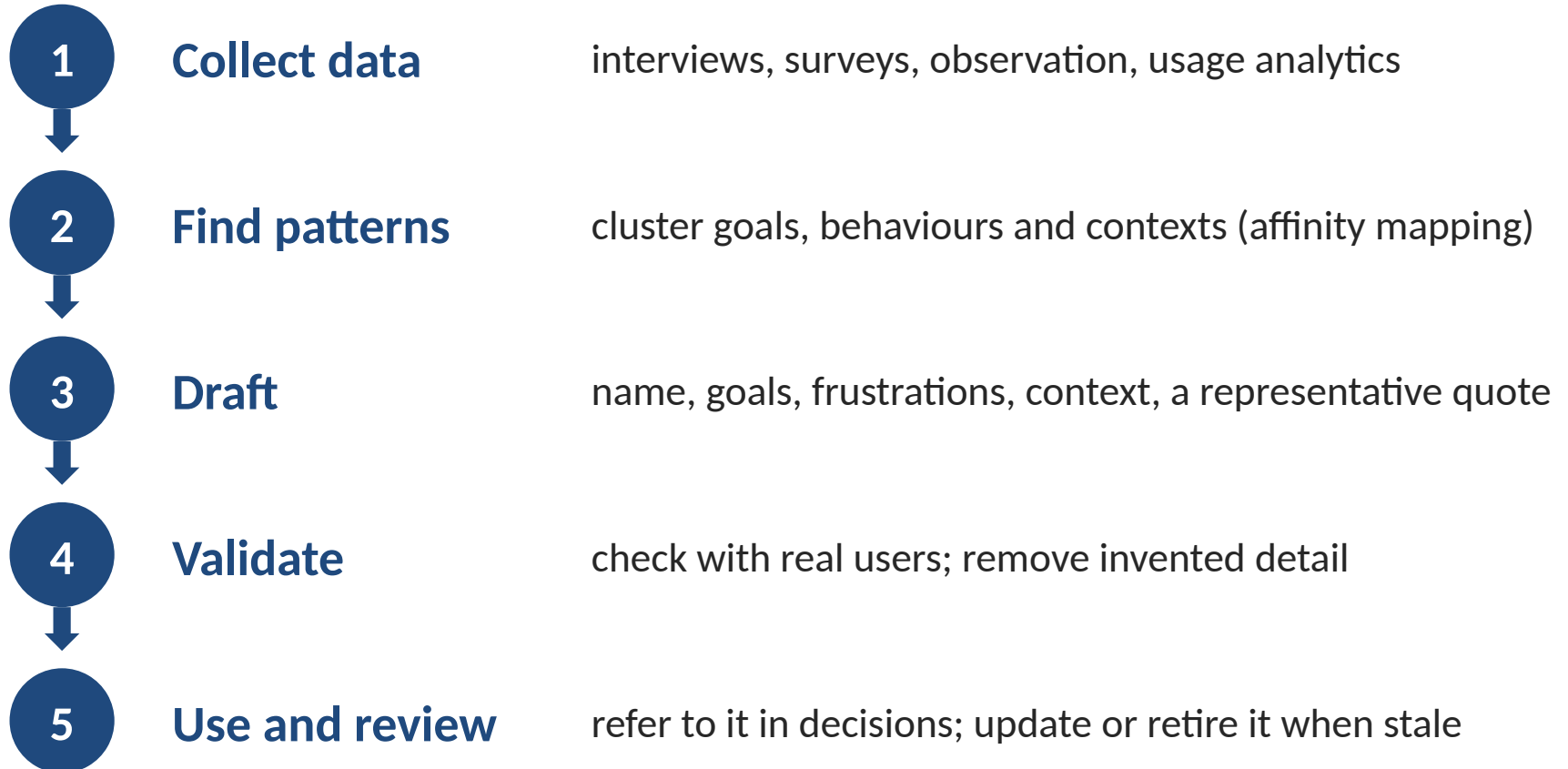
Keep users in view

Align the team

Test decisions: "Would Priya manage this?"

A persona is not an average user and not a stereotype: every attribute should trace back to research data.

Building Personas from Research Data



Evidence first: if you cannot point to the data behind an attribute, leave it out.

Example Persona: Priya



Priya, 20

2nd-year engineering student

- Screen reader user
- Commutes 30 minutes
- Studies in 2 to 3 hour blocks

Goals

- Find a quiet, step-free room without wasting a journey.
- Book by keyboard and screen reader in under two minutes.

Frustrations

- Dashboards that show room status only as colours or icons.
- Rooms that are full or noisy on arrival.

"I just want to know a room is free before I leave home."

Illustrative fictional persona, based on 8 interviews and 120 survey responses (numbers invented for the example).

Integrating Personas and Use Cases

Use case	Priya	Sam	Alex
Book a study room	●	●	-
Cancel a booking	○	●	-
Check availability before travelling	●	●	○
Report a faulty sensor	-	○	●
Get an audible "room free" alert	●	Conflict	-

● primary ○ occasional - not needed. Sam: commuter. Alex: facilities technician.

- An empty row: a use case nobody needs. An empty column: a persona nobody serves.
- A conflict (Priya needs audio alerts; Sam studies in a silent library) is a design decision to make early.
- A scenario tells one persona's use case as a short story in context.

Strengths and Limitations

Strengths

- Build empathy and a shared picture of users.
- Make behaviour concrete and testable.
- Expose coverage gaps and conflicting needs.
- Bridge business and technical audiences.

Limitations

- Bias and stereotype drift from invented detail.
- Stale specifications if never reviewed.
- Cost and over-documentation.
- Typical users are covered; edge cases and inclusion can be missed.

Right-size the documentation: lightweight for agile iterations, richer artefacts for regulated or high-risk domains. Plan reviews to keep insights current.

Summary and What's Next

- **Use cases:** capture behaviour at the system boundary
- **Personas:** anchor it in evidence about real users
- **Levels:** specify at user-goal level, with alternative flows
- **Integration:** map personas to use cases to find gaps and conflicts
- **Care:** watch for bias, stereotypes and stale documents

Let's read and explore

- Cockburn, A. (2001). Writing Effective Use Cases. Addison-Wesley.
- Jacobson, I. et al. (1992). Object-Oriented Software Engineering: A Use Case Driven Approach.
- Cooper, A. (1999). The Inmates Are Running the Asylum.
- Pruitt, J. and Adlin, T. (2006). The Persona Lifecycle.
- ISO 9241-210:2019. Human-centred design for interactive systems.

Next: System architecture: turning requirements into components and interfaces.